

CALIBRATION SENSITIVITY IN MINIMAX LOW-RANK APPROXIMATION

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ABSTRACT

A natural way to merge several low-rank task updates of one linear layer is to choose a single rank- r update that minimises the largest per-task functional error measured on calibration activations. We study how sensitive this empirical minimax problem is to the calibration sample, in a controlled synthetic layer-level setting with 4 tasks, input dimension 16, output dimension 12, rank budget $r = 4$ and 32 calibration inputs per task. We first check the optimiser. Across 39 minimax fits, the relative gap between a Lagrangian lower bound, computed with an exactly solvable weighted reduced-rank regression, and the achieved worst-task value is at most 2.0×10^{-9} , and an independent raw-data reference agrees to within 2.5×10^{-15} . These floating-point checks argue against large optimisation error on these instances; they are not certificates. The optimised calibration objective is nevertheless optimistic: on fresh calibration draws the minimax solution’s fitted worst-task error (0.497 on average) understates its population value (0.672), and its population worst-task error is not lower than that of the uniform-weight solution on average (mean paired difference +0.007 over 3 problem instances $\times$ 3 nested calibration draws; the pre-registered verdict is ambiguous). Replacing the empirical second moments by population moments, an oracle that exists only in simulation, removes the average disadvantage, whereas replacing only the error normalisers does not. Under our pre-registered attribution rule the cause remains undetermined. Elementary, known perturbation bounds that make the dependence on moment error explicit hold on every recorded fit, but at this sample size they are loose by a factor of at least 61. No real adapters were evaluated.

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1 INTRODUCTION

Low-rank adapters (Hu et al., 2021) make it cheap to specialise one base model to many tasks, and a growing literature merges several such adapters into a single update without retraining (Ilharco et al., 2022; Yadav et al., 2023; Jin et al., 2022; Stoica et al., 2024; He et al., 2026). Two design choices recur. First, some methods measure the discrepancy between the merged and the task-specific layers on calibration activations rather than in weight space (Jin et al., 2022; Nguyen et al., 2025). Second, some fix the rank of the merged update in advance (He et al., 2026; Amballa et al., 2026). Merging can also hurt individual tasks badly (Cao et al., 2026), and worst-task retention is reported as an evaluation metric (He et al., 2026). Taken together, these choices suggest a layer-level objective: choose one rank- r update that minimises the largest per-task functional error on calibration data. With identity targets this is exactly min–max reconstruction fair PCA (Samadi et al., 2018; Tantipongpipat et al., 2019). With fixed task weights it reduces to classical reduced-rank regression (Izenman, 1975).

This paper is about one uncomfortable observation made while instrumenting that objective. In a synthetic layer-level fixture with three problem instances, the minimax solution had a lower calibration worst-task error than the solution with uniform task weights (0.472 versus 0.601 on average) but a higher held-out worst-task error (0.857 versus 0.685). Before this can inform a method, one has

to know why. The candidates are that the solver failed, that the calibration second moments are too noisy, that the per-task normalisers in the relative error are too noisy, or that the evaluated objective differs from the optimised one.

We therefore run a diagnostic study rather than propose a method. Our findings are the following.

- **Optimisation check.** Every minimax fit is bracketed by a Lagrangian lower bound, which is computed with an exactly solvable inner problem, and by the achieved worst-task value. The relative gaps are at most 2.0×10^{-9} over 39 fits. An independent reduced-rank regression computed from raw samples reproduces the inner values to 2.5×10^{-15} . On these instances this argues against large optimisation error. It is a floating-point check, not a proof that the implementation is free of bugs or that the solver is globally optimal in general.
- **Calibration optimism.** On 9 fresh calibration draws, nested in 3 problem instances, the minimax solution’s fitted worst-task error understates its population value by +0.175 on average. For the uniform-weight solution the understatement is +0.060. On these draws the minimax solution is not better than the uniform one on the population worst-task error (mean paired difference +0.007; the pre-registered verdict is ambiguous). The larger gap first observed came from one particular calibration draw, not from an evaluation artefact.
- **Oracle substitution.** In a pre-registered 2×2 design, replacing the empirical second moments by their population values removes the average disadvantage. Replacing only the normalisers does not. The pattern is consistent with moment-estimation noise, but the pre-registered attribution rule is not met, so the cause remains undetermined.
- **Elementary bounds.** We record standard perturbation and uniform-deviation bounds, including the case of estimated normalisers. They make the dependence of the excess population error on $\|S_k - \hat{S}_k\|_{\text{op}}$ explicit. We claim no novelty for them. On the recorded fits they hold but are loose by a factor of at least 61.

Scope. All results are on a synthetic linear layer whose population moments are known in closed form. The objective is layer-local; it is not an end-to-end task loss. We evaluated no real adapters. Four public candidates were excluded because their usage terms, their base-model revision and their checkpoint-selection split could not be established (Appendix F). We make no claim about merging accuracy, and we do not claim novelty for the solver components (Section 2).

2 RELATED WORK

Merging task updates. Task arithmetic adds scaled task vectors (Ilharco et al., 2022), and TIES-Merging resolves sign and magnitude conflicts (Yadav et al., 2023). RegMean minimises the summed squared difference of layer outputs on calibration inputs in closed form (Jin et al., 2022). RegMean++ changes where those inputs come from (Nguyen et al., 2025). Several methods for LoRA adapters work with the effective update $B_k A_k$ rather than with its factors:

- KnOTS aligns updates through a joint SVD (Stoica et al., 2024);
- Core Space merges in a lossless reduced basis (Panariello et al., 2025);
- Compress-then-Merge produces a rank- r adapter by construction from weights alone (He et al., 2026);
- budget-aware selection allocates a global rank budget with a data-free proxy (Amballa et al., 2026).

Singular-vector methods address interference between tasks (Gargiulo et al., 2024; Marczak et al., 2025). Calibration-aware low-rank variants exist in partial forms: LoRM fixes one factor (Salami et al., 2024), ESM truncates each task separately (Li et al., 2026), and IterIS uses per-task weights (Chen et al., 2024). A related line compresses many adapters for serving without merging them (Brüel-Gabrielsson et al., 2024). None of the works we checked optimise a max-over-tasks functional error under a rank constraint. In our fixed-weight baseline, the $S_\lambda^{1/2}$ -whitened truncation is a direct application of reduced-rank regression, and we do not claim it as a contribution.

Multi-objective and worst-case merging. Pareto Merging learns preference-conditioned merges with a smooth Tchebycheff scalarisation (Chen & Kwok, 2024), and MAP approximates Pareto fronts over merging coefficients (Li et al., 2024). Cao et al. (2026) study task-level merging collapse and give a rate–distortion bound on worst-case hidden-state distortion. For linear models, functional distortion measured with the input second moment is the distortion used in rate–distortion analyses of compression (Gao et al., 2018).

Minimax low-rank approximation. With identity targets our objective is min–max reconstruction fair PCA. The relevant known results are:

- a semidefinite relaxation that solves the two-group case exactly and has a gap for three groups (Samadi et al., 2018; Tantipongpipat et al., 2019);
- hardness of approximating socially fair low-rank approximation (Song et al., 2024);
- distributionally robust and worst-case PCA (Wang et al., 2025; Fries et al., 2026).

The multiplicative-weights scheme with an eigenvector inner step in Samadi et al. (2018), and the exponentiated group weights of Group DRO (Sagawa et al., 2019), have the same structure as our dual ascent. The fixed-weight subproblem is reduced-rank regression (Izenman, 1975), solved by an Eckart–Young truncation (Eckart & Young, 1936) in a whitened metric. General elementwise-weighted low-rank approximation has no such closed form (Srebro & Jaakkola, 2003). Zero duality gap without a rank constraint follows from Sion’s minimax theorem (Sion, 1958).

3 PROBLEM AND EXPERIMENTAL SETTING

Objective. Consider one linear layer and K tasks. Task k has an adapter with effective update $D_k = \frac{\alpha}{r_0} B_k A_k \in \mathbb{R}^{p \times q}$. We only ever use D_k and never the factors, so every quantity below is invariant under $(B_k, A_k) \mapsto (B_k R, R^{-1} A_k)$. Inputs of task k are $x \in \mathbb{R}^q$ with raw second moment $S_k = \mathbb{E}[xx^\top]$, which equals the covariance only when $\mathbb{E}[x] = 0$. For a merged update $W \in \mathbb{R}^{p \times q}$, a positive semidefinite S and a normaliser $d > 0$ we write

$$E_k(W; S, d) = \frac{\text{tr}[(W - D_k) S (W - D_k)^\top]}{d}. \quad (1)$$

With $S = S_k$, the numerator is the expected squared change of the layer output on task k when D_k is replaced by W . We use relative errors, in which $d_k = \text{tr}(D_k S_k D_k^\top)$ is the error of dropping the adapter. The population and calibration worst-task objectives are

$$F(W) = \max_k E_k(W; S_k, d_k), \quad \hat{F}(W) = \max_k E_k(W; \hat{S}_k, \hat{d}_k), \quad (2)$$

where $\hat{S}_k = X_k X_k^\top / n$ is computed from n calibration inputs and $\hat{d}_k = \text{tr}(D_k \hat{S}_k D_k^\top)$. The feasible set is $\mathcal{W}_r = \{W : \text{rank}(W) \leq r\}$. It is not bounded: nothing restricts $\|W\|_F$.

Methods compared. Both methods return a rank- r update and use the same calibration inputs with task identity. Neither has tuned hyper-parameters.

- *Uniform WRRR* minimises $\frac{1}{K} \sum_k E_k(W; \hat{S}_k, \hat{d}_k)$ over $\mathcal{W}_r$. This is a weighted reduced-rank regression with a closed-form solution (Section 4).
- *Minimax* approximately minimises $\hat{F}$ over $\mathcal{W}_r$, using Lagrangian dual ascent with exact inner solves followed by monotone primal refinement.

We report both the lower bound (LB) and the achieved value (UB) for every minimax fit.

Synthetic layer. The generator is fixed by the configuration of the recorded runs (Appendix G): $p = 12$, $q = 16$, $K = 4$ tasks, adapter rank $r_0 = 4$ and merged rank $r = 4$. Adapter factors mix a shared component (correlation 0.5) with a task-specific component, and task gains are $(1, 1, 0.6, 1.6)$. Inputs are $x = L_k z$ with $z \sim \mathcal{N}(0, I)$, where L_k is a random rotation times a diagonal with decay $0.8^{i/2}$. Hence $\mathbb{E}[x] = 0$ and $S_k = L_k L_k^\top$ exactly. Each task has $n = 32$ calibration, 64 development and 512 test inputs. Because S_k is known, F can be evaluated exactly as $\|(W - D_k) L_k\|_F^2 / \|D_k L_k\|_F^2$. A Monte Carlo check with 20000 samples per task confirmed the closed form for all 12 task moments; the largest standardised entrywise deviation was 3.38.

Design. A *teacher* is one problem instance: its adapters and input distributions. Stage 1 ran both methods on 3 teachers with one calibration draw each and evaluated them on the test sample. Stage 2 was pre-registered before execution (commit 06947aef). For the same 3 teachers it drew 3 fresh calibration sets per teacher, giving 9 cells nested in teachers. We do not count them as 9 independent experiments. Each fit is run in four arms:

- *Emp/Emp*: empirical moments and empirical normalisers (the deployable setting);
- *Emp/Pop* and *Pop/Emp*: one of the two quantities is replaced by its population value;
- *Pop/Pop*: both are population values.

Pop/Pop uses no calibration sample and is computed once per teacher. The original, previously seen calibration draw is reported separately. Every solution is evaluated on the same population objective F . The pre-registered decision rules use the paired difference $\Delta = F(\hat{W}_{\text{minimax}}) - F(\hat{W}_{\text{uniform}})$:

- *problem present*: mean Δ over the new cells exceeds +0.02 and the teacher-level mean Δ is positive for at least two teachers;
- *problem absent*: mean $\Delta \leq 0$;
- *ambiguous*: otherwise.

Moment noise would be attributed only if the problem were present in both empirical-moment arms and absent in both population-moment arms.

4 ANALYSIS OF THE METHOD UNDER STUDY

This section collects what is known about the optimisation problem and adds elementary sensitivity bounds. Proofs are in Appendix A. None of the statements is new; we state them in our notation so that the experiments can be read against them.

4.1 FIXED TASK WEIGHTS: REDUCED-RANK REGRESSION

For λ in the simplex Δ_K , the weighted objective is $L_\lambda(W) = \sum_k \lambda_k E_k(W; S_k, d_k) = \text{tr}(W A_\lambda W^\top) - 2 \text{tr}(W C_\lambda^\top) + c_\lambda$, where $A_\lambda = \sum_k (\lambda_k/d_k) S_k$ and $C_\lambda = \sum_k (\lambda_k/d_k) D_k S_k$. The output side carries the identity weight, so the problem is reduced-rank regression (Izenman, 1975).

Proposition 1 (known). *If $A_\lambda \succ 0$, then $W_\lambda = [C_\lambda A_\lambda^{-1/2}]_r A_\lambda^{-1/2}$ minimises L_λ over $\mathcal{W}_r$. Here $[\cdot]_r$ is a best rank- r approximation in Frobenius norm. The minimiser is unique if and only if $\sigma_r > \sigma_{r+1}$ for the singular values of $C_\lambda A_\lambda^{-1/2}$. For $r \geq \min(p, q)$ it is the λ -weighted RegMean solution $C_\lambda A_\lambda^{-1}$.*

Positive definiteness of A_λ alone does not give uniqueness; the singular-value gap at the truncation boundary is also needed. If A_λ is singular, the objective does not identify W on $\ker A_\lambda$. Our implementation then uses a pseudo-inverse and returns the minimiser that vanishes on $\ker A_\lambda$. A ridge term changes the objective, so its value is never used as a bound for the original problem.

4.2 WORST-TASK OBJECTIVE AND A LAGRANGIAN BOUND

Let $\text{OPT} = \min_{W \in \mathcal{W}_r} F(W)$, and define the same quantities for $\hat{F}$. Since $\max_k a_k = \max_{\lambda \in \Delta_K} \sum_k \lambda_k a_k$, weak duality gives

$$g(\lambda) = \min_{W \in \mathcal{W}_r} L_\lambda(W) \leq \text{OPT} \leq F(W) \quad \text{for all } \lambda \in \Delta_K \text{ and } W \in \mathcal{W}_r. \quad (3)$$

Here g is concave, and $(E_k(W_\lambda))_k$ is a supergradient at λ . By Proposition 1, $g(\lambda)$ can be evaluated exactly, so exponentiated-gradient ascent on λ yields certified lower bounds in exact arithmetic. Any feasible W gives an upper bound.

The rank constraint makes the problem non-convex, and the gap can be strictly positive. With $D_k = I$ the problem is min-max reconstruction fair PCA. There g equals a semidefinite relaxation

value, and a published three-group example has relaxation value $7/4$ against an exact value of $26/17$ (Tantipongpipat et al., 2019, Lemma 6.2). Our test suite reproduces both values. Without the rank constraint the gap is zero by Sion’s theorem (Sion, 1958).

Proposition 2 (standard). *If $g(\lambda^*) = \text{OPT}$ and L_{λ^*} has a unique minimiser W_{λ^*} over $\mathcal{W}_r$, then W_{λ^*} is the unique minimiser of F over $\mathcal{W}_r$.*

In floating point we observe only a small gap and an estimated λ^* . Propositions 1–2 therefore interpret our numerical checks but do not certify them rigorously.

4.3 SENSITIVITY TO THE CALIBRATION MOMENTS

Throughout, $\Delta_k = S_k - \hat{S}_k$ is symmetric.

Lemma 1 (fixed normaliser; spectral-norm inequality). *For any W and $d > 0$, $|E_k(W; S_k, d) - E_k(W; \hat{S}_k, d)| \leq \|\Delta_k\|_{\text{op}} \|W - D_k\|_F^2 / d$.*

Lemma 2 (uniform deviation; standard). *Let $\mathcal{W}$ be any set and suppose $\sup_{W \in \mathcal{W}} |F(W) - \hat{F}(W)| \leq \delta$. If $\hat{W} \in \mathcal{W}$ satisfies $\hat{F}(\hat{W}) \leq \inf_{\mathcal{W}} \hat{F} + \epsilon_{\text{opt}}$, then $F(\hat{W}) - \inf_{\mathcal{W}} F \leq 2\delta + \epsilon_{\text{opt}}$.*

The rank constraint alone does not bound $\|W\|_F$. We therefore apply the lemmas to the analysis set $\mathcal{W}_{r,B} = \{W \in \mathcal{W}_r : \|W\|_F \leq B\}$. Our solver optimises over $\mathcal{W}_r$ and never enforces such a bound.

Corollary 1. *With fixed normalisers d_k (the Emp/Pop arm), $\sup_{\mathcal{W}_{r,B}} |F - \hat{F}| \leq \delta_{\text{fix}} := \max_k \|\Delta_k\|_{\text{op}} (B + \|D_k\|_F)^2 / d_k$. If $\hat{W} \in \mathcal{W}_{r,B}$ minimises $\hat{F}$ over $\mathcal{W}_r$ up to ϵ_{opt} and $W^* \in \mathcal{W}_{r,B}$, then $F(\hat{W}) - F(W^*) \leq 2\delta_{\text{fix}} + \epsilon_{\text{opt}}$.*

Lemma 3 (estimated normalisers). *Assume $d_k > 0$ and $\hat{d}_k \geq d_{\min,k} > 0$. Then for any W ,*

$$|E_k(W; S_k, d_k) - E_k(W; \hat{S}_k, \hat{d}_k)| \leq \frac{\|\Delta_k\|_{\text{op}}}{\hat{d}_k} \left(\|W - D_k\|_F^2 + E_k(W; S_k, d_k) \|D_k\|_F^2 \right).$$

The first term is the moment error. The second is the normaliser error, which uses $|\hat{d}_k - d_k| \leq \|\Delta_k\|_{\text{op}} \|D_k\|_F^2$.

These are deterministic inequalities. We make no probabilistic statement about $\|\Delta_k\|_{\text{op}}$ at our sample size, and we do not infer a rate from the observations. Section 5 checks the inequalities numerically on the recorded instances and reports how loose they are.

5 EXPERIMENTS AND OBSERVATIONS

All numbers below are generated from the recorded result files by one exporter script (Appendix G). Table 6 in the appendix lists all stage-1 methods, including the weight-space baselines, which are not discussed further here.

5.1 THE ORIGINAL OBSERVATION AND WHAT ITS NUMBERS COMPARE

Stage 1 reported mean test worst-task errors of 0.857 for minimax and 0.685 for uniform WRRR. Table 1 separates the comparisons that were later conflated in a summary. The triple 0.713/0.802/1.057 is the *minimax* solution on the *test sample*. The triple 0.726/0.768/1.081 is the *same* solution under the *population* objective. Sample and population agree closely, so the original gap is not an artefact of the test sample. On this draw the minimax solution is worse than the uniform solution for every teacher under both evaluations. It is also the solution with the lowest calibration objective in every column; for minimax the calibration errors of all tasks are equalised at the value shown in the “calibration $\hat{F}$ ” row.

5.2 IS THE OPTIMISER THE PROBLEM?

Table 2 summarises the audit.

Table 1: The original calibration draw of each teacher, stage 1 and stage 2. Each column is one teacher. The same fitted matrix is evaluated three ways: the test sample (seen during development), the exact population objective F , and the calibration objective $\hat{F}$ it was fitted on. The stage-1 and stage-2 fits are bit-identical (matching SHA-256 of the merged matrices).

Solution	Evaluation	Teacher 0	Teacher 1	Teacher 2
Minimax	test sample ($n=512$, seen)	0.713	0.802	1.057
Minimax	population F	0.726	0.768	1.081
Minimax	calibration $\hat{F}$	0.449	0.463	0.503
Uniform WRRR	test sample ($n=512$, seen)	0.642	0.660	0.754
Uniform WRRR	population F	0.657	0.655	0.780
Uniform WRRR	calibration $\hat{F}$	0.536	0.624	0.643

Table 2: Numerical audit of the 39 minimax fits in stage 2, by arm. The relative gap is $(UB - LB)/UB$ on the fitted objective. The second column of numbers compares the lower bound with an independent reduced-rank regression computed from raw samples at the same λ^* . The last three columns are the smallest task weight, the largest condition number of A_{λ^*} , and the smallest relative singular-value gap at the truncation boundary. Floating-point values, not certificates.

Moment / normaliser	Fits	max rel. gap	max $ g - g_{\text{ref}} /g_{\text{ref}}$	min λ_k	max $\kappa(S_\lambda)$	min $(\sigma_r - \sigma_{r+1})/\sigma_r$
Emp/Emp	12	7.4×10^{-10}	2.5×10^{-15}	0.136	19.3	0.109
Emp/Pop	12	5.8×10^{-10}	2.5×10^{-15}	0.130	16.3	0.041
Pop/Emp	12	2.0×10^{-9}	1.3×10^{-15}	0.170	7.9	0.011
Pop/Pop	3	5.9×10^{-10}	1.1×10^{-15}	0.165	6.1	0.014

- Every fit closes its dual gap to at most 2.0×10^{-9} in relative terms. In 9 fits the primal refinement stage was needed.
- The lower bounds agree with the independent raw-data reference to within 2.5×10^{-15} .
- Every achieved upper bound agrees with a raw-sample re-evaluation to within 2.8×10^{-15} .
- All solutions satisfy the rank constraint: $\sigma_{r+1}/\sigma_1 \leq 2.0 \times 10^{-16}$.
- All λ^* lie in the interior of the simplex (smallest weight 0.130).
- Each A_{λ^*} is well conditioned (condition number at most 19.3).
- The smallest relative singular-value gap at the truncation boundary is 0.011.

This gap is small for some Pop/Emp fits but nonzero, so by Proposition 1 the inner minimiser at λ^* is unique on every instance. On these instances a large optimisation error therefore does not explain the calibration–population discrepancy. This rules out neither every possible implementation bug nor failure on other problems.

5.3 ORACLE SUBSTITUTION: MOMENTS VERSUS NORMALISERS

Table 3 and Figure 1 give the four arms. The pre-registered verdicts are:

- Emp/Emp: AMBIGUOUS (mean Δ +0.007, sd over cells 0.078);
- Emp/Pop: PROBLEM_PRESENT (+0.039);
- Pop/Emp: PROBLEM_ABSENT (−0.025);
- Pop/Pop: PROBLEM_ABSENT (−0.093).

Three observations follow.

First, the large disadvantage seen on the original draw does not reproduce on average over fresh draws. The original draw gave +0.069, +0.113 and +0.301 for teachers 0, 1 and 2. The fresh draws give teacher means −0.041, −0.014 and +0.075. The draw-to-draw spread is of the same order as the effect.

Table 3: Paired difference $\Delta = F(\hat{W}_{\text{minimax}}) - F(\hat{W}_{\text{uniform}})$ on the population objective; negative values favour minimax. The teacher columns average the 3 fresh draws of that teacher, and the last column averages all 9 nested cells. Pop/Pop does not depend on the draw and is computed once per teacher.

Moment / normaliser	Teacher 0	Teacher 1	Teacher 2	Mean over 9 cells (sd)
Emp/Emp	−0.041	−0.014	+0.075	+0.007 (0.078)
Emp/Pop	+0.034	−0.002	+0.086	+0.039 (0.082)
Pop/Emp	−0.041	−0.025	−0.010	−0.025 (0.043)
Pop/Pop	−0.066	−0.137	−0.076	−0.093 (−)

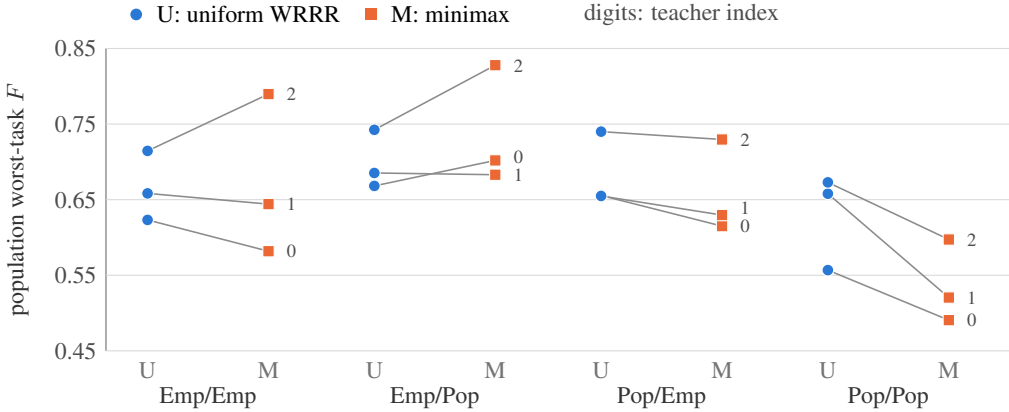


Figure 1: Teacher-level paired comparison on the population worst-task error F . For each arm, the U (uniform WRRR, circle) and M (minimax, square) values of each teacher are joined by a line, using means over the 3 fresh draws (Pop/Pop does not depend on the draw). Lower is better. Per-draw values are in Table 5.

Second, substituting population moments makes minimax better than uniform on average for both choices of normaliser. Substituting population normalisers alone does not: Emp/Pop is worse than Emp/Emp. One explanation we have *not* tested is that, when both come from the same draw, the moment error and the normaliser error partly cancel in the ratio.

Third, in the Pop/Pop arm the fitted objective *is* the evaluation objective. That minimax wins there (for teachers 0, 1 and 2, F is 0.491, 0.520 and 0.597 against 0.557, 0.658 and 0.673) is a consequence of optimising the right objective. It is not evidence that minimax generalises. It does show that the evaluator and the fitted objective are consistent.

The pattern is consistent with moment-estimation noise driving the calibration–population discrepancy. The pre-registered attribution rule, however, required the problem to be *present* in both empirical-moment arms, and Emp/Emp is ambiguous. The formal outcome is therefore **cause undetermined**. We report the moment-noise reading as a hypothesis supported by the oracle substitution, not as an established cause.

5.4 CALIBRATION OPTIMISM IS DISTINCT FROM THE NUMERICAL GAP

Table 4 shows that both solutions look better on calibration data than they are, and that the minimax solution is more optimistic in every arm that uses a calibration sample. For example, in Emp/Emp the minimax understatement is +0.175, against +0.060 for uniform. This fits the familiar picture that minimising a maximum of noisy estimates selects the noise. At 32 samples per task the relative normaliser estimates range from 0.72 to 1.73 times their population values.

Table 4: Fitted objective of the returned solution against its population value, averaged over the 9 fresh cells. This is a statistical optimism gap. It is unrelated to the numerical UB–LB gap of Table 2, which is at most 2.0×10^{-9} in relative terms.

Moment / normaliser	Solution	Fitted objective $\hat{F}(\hat{W})$	Population $F(\hat{W})$	Difference
Emp/Emp	Uniform WRRR	0.606	0.665	+0.060
Emp/Emp	Minimax	0.497	0.672	+0.175
Emp/Pop	Uniform WRRR	0.622	0.699	+0.076
Emp/Pop	Minimax	0.517	0.738	+0.220
Pop/Emp	Uniform WRRR	0.612	0.684	+0.072
Pop/Emp	Minimax	0.513	0.658	+0.145

5.5 DO THE DETERMINISTIC BOUNDS EXPLAIN THE GAP?

We evaluated Corollary 1 (Emp/Pop) and Lemmas 3 and 2 (Emp/Emp) on all 24 recorded fits. The setup was as follows:

- W^* is the Pop/Pop solution of the same teacher;
- B is the larger of $\|\hat{W}\|_F$ and $\|W^*\|_F$, chosen after the fact, which is legitimate because the inequalities are deterministic (median 5.274);
- ϵ_{opt} is the fit’s own UB–LB gap (at most 3.8×10^{-10}).

The fitted matrices had been stored only as hashes. They were regenerated with the identical code path, and all 27 regenerated matrices match their recorded SHA-256 (Appendix D).

Every inequality holds: 24 of 24 excess bounds and 192 of 192 pointwise checks of Lemmas 1 and 3. None of them is informative at this sample size. The relative moment error $\max_k \|S_k - \hat{S}_k\|_{\text{op}} / \|S_k\|_{\text{op}}$ lies between 0.64 and 1.04. The smallest excess bound is 14.026 in units of relative error, at least 61 times the realised excess $F(\hat{W}) - F(W^*)$. Even the pointwise inequalities use at most a fraction 0.27 of their right-hand side. The bounds are therefore consistent with the observations but explain neither their size nor their sign. With $n = 32$ samples in dimension $q = 16$, a worst-case spectral argument is far from tight for these fits.

6 DISCUSSION AND LIMITATIONS

The uniform-weight baseline is hard to beat here. Uniform WRRR has a closed form and does not concentrate on the task that looks worst on the calibration sample. On fresh draws it was not worse than the empirical minimax solution on average, and its calibration optimism was smaller (Table 4). We therefore keep it as the primary baseline for any further work. A worst-task objective would have to beat it at the same rank and the same calibration access before it is worth its cost. In stage 2 a minimax fit took a median of 3.7 s of CPU time (at most 19.8 s, including the bound audit), against a median of 8 ms for the closed form.

What we could not separate.

- *Relative normalisation.* Uniform WRRR and minimax both use relative errors. The relative-error choice is itself a design decision, and we did not compare it with absolute errors in this study. Its effect is not separated from the effects reported here.
- *Moment–normaliser interaction.* The Emp/Pop arm was worse than Emp/Emp. A plausible reading is that errors in $\hat{S}_k$ and in $\hat{d}_k$, which are computed from the same draw, partly cancel in the ratio. We did not test this.
- *Cause.* The pre-registered rule leaves the cause undetermined. The oracle arms tell us what changes when an input to the objective is replaced by its population value. They do not identify a mechanism, and population moments are not available to a deployable method.

Statistical limitations. The design has 3 teachers and 3 calibration draws nested in each teacher. We report means, spreads and per-draw values (Table 5) and run no significance test. A non-significant or ambiguous difference is not evidence that the two methods are equivalent. The original draw was seen before the diagnostic was designed, so stage 2 is a development diagnostic, not a confirmatory experiment.

Scope of the setting. The objective is the error of one linear layer’s output. It is not an end-to-end task loss, and gains in it need not transfer to accuracy. The synthetic generator uses Gaussian inputs with a known second moment and a single configuration of dimension, rank and sample size. We ran no sweep over these quantities. The bounds of Section 4 are deterministic and hold on a norm-bounded analysis set, whereas the solver optimises over the unbounded rank-constrained set.

Real adapters. We checked 4 public LoRA adapters (Flan-T5-base, GLUE CoLA, MRPC, RTE and SST-2; Chung et al., 2022; Tang et al., 2024) using metadata only. They were excluded for three reasons:

- the adapter repositories carry no license information;
- the base-model revision used for training is not recorded;
- the training and checkpoint-selection split is not documented.

The publisher evaluates on the GLUE validation split, which is the split we had planned to use for development and testing. The publisher also reports exact-match accuracy of generated label words for all four tasks. The official GLUE metrics differ for two of them: Matthews correlation for CoLA and accuracy/F1 for MRPC (Wang et al., 2018). These scales should not be mixed into a single percentage-point gain. Appendix F lists the planned but unexecuted real-adapter pilot.

7 CONCLUSION

We instrumented a worst-task, rank-constrained merging objective for linear-layer updates and studied why its empirical minimiser looked better on calibration data than on held-out data. On the synthetic instances we examined, the optimiser was not the problem: Lagrangian lower bounds, achieved values and an independent reference agree to floating-point precision. The fitted worst-task objective is nevertheless optimistic, more so for minimax than for uniform task weights. On fresh calibration draws minimax did not improve the population worst-task error on average. Replacing empirical second moments by population moments removed the disadvantage, and replacing the normalisers alone did not. Our pre-registered rule nonetheless leaves the cause undetermined. Standard deviation bounds hold on every fit but are far too loose at this sample size to account for the observed gaps. The practical conclusion is modest: at this calibration size a closed-form uniform-weight solution remains the baseline to beat. Whether any of this carries over to real adapters and task accuracy is open; we did not test it.

AI USE STATEMENT

This draft was produced almost entirely by a generative AI system: Claude, an AI assistant made by Anthropic, working as a coding and writing agent in a repository session. No human author has yet reviewed it (status: HUMAN_REVIEW_PENDING). The disclosure categories of the ICLR 2027 AI Policy for Authors apply as follows.

- *Required disclosures.*
 - The AI system implemented all methods, references and experiment runners.
 - It generated the synthetic data sets.
 - It formulated the hypotheses and designed the diagnostic experiments.
 - It formulated the mathematical statements and wrote their proofs, using standard arguments.
 - It interpreted the results.

The research question and the constraints on scope, compute and permitted assets were set by a human in natural-language instructions.

- *Recommended disclosures.* The AI system also:
 - drafted every section of this paper;
 - created the tables and the figure;
 - searched for and summarised related literature, partly through AI sub-agents;
 - formatted the references from metadata fetched from arXiv, Crossref and the AAAI proceedings.

Verification was also done by the AI system, and is limited to the following:

- automated unit tests and independent numerical references (Appendix G);
- citation metadata that is machine-extracted rather than typed;
- a claim–evidence table linking statements to files (`paper/claim_evidence.tsv`).

Most related-work descriptions rest on summaries by AI sub-agents, and only a few sources were read directly (see the claim–evidence table). No human has yet checked the proofs, the literature summaries or the code. Responsibility for the final content lies with the human authors once they have reviewed it.

REPRODUCIBILITY STATEMENT

Appendix G lists the exact commands, the commit identifiers, the configuration and data hashes, and the resource limits of every run. All tables and numbers in the text are generated from the recorded result files by one exporter script. Appendix A states the assumptions and gives complete proofs of the elementary lemmas. Appendix F lists every experiment that was planned but not run.

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626 A PROOFS

627 The statements below are standard. We include proofs so that the assumptions are explicit.
 628

629 **Proposition 1.** Write $L_\lambda(W) = \|(W - M_\lambda)A_\lambda^{1/2}\|_F^2 + \text{const}$ with $M_\lambda = C_\lambda A_\lambda^{-1}$. This is valid
 630 when $A_\lambda \succ 0$ and follows by completing the square in $\text{tr}(W A_\lambda W^\top) - 2 \text{tr}(W C_\lambda^\top)$. The change
 631 of variables $N = W A_\lambda^{1/2}$ is a bijection of $\mathcal{W}_r$ onto itself, because $A_\lambda^{1/2}$ is invertible and preserves
 632 rank. Minimising $\|N - C_\lambda A_\lambda^{-1/2}\|_F^2$ over $\text{rank}(N) \leq r$ is the Eckart–Young problem (Eckart &
 633 Young, 1936). Its minimisers are the rank- r truncated SVDs, and they are unique if and only if
 634 $\sigma_r > \sigma_{r+1}$. Mapping back gives W_λ . For $r \geq \min(p, q)$ no truncation occurs and $W_\lambda = M_\lambda$. $\square$
 635

636 **Proposition 2.** Let W° minimise F over $\mathcal{W}_r$. Then $\text{OPT} = F(W^\circ) \geq \sum_k \lambda_k^* E_k(W^\circ) =$
 637 $L_{\lambda^*}(W^\circ) \geq g(\lambda^*) = \text{OPT}$. Hence W° minimises L_{λ^*} , and uniqueness gives $W^\circ = W_{\lambda^*}$. $\square$
 638

639 **Lemma 1.** Let $Y = W - D_k$ with rows $y_i^\top$. Then $\text{tr}(Y \Delta_k Y^\top) = \sum_i y_i^\top \Delta_k y_i$. For symmetric
 640 Δ_k , $|y^\top \Delta_k y| \leq \|\Delta_k\|_{\text{op}} \|y\|^2$. Summing over rows gives $|\text{tr}(Y \Delta_k Y^\top)| \leq \|\Delta_k\|_{\text{op}} \|Y\|_F^2$, and
 641 dividing by $d > 0$ proves the claim. This is the usual spectral-norm bound on a quadratic form. $\square$
 642

643 **Lemma 2.** For any $W' \in \mathcal{W}$, $F(\hat{W}) \leq \hat{F}(\hat{W}) + \delta \leq \hat{F}(W') + \epsilon_{\text{opt}} + \delta \leq F(W') + 2\delta + \epsilon_{\text{opt}}$.
 644 Taking the infimum over W' gives the claim. This is the standard uniform-deviation argument for
 645 approximate empirical minimisers. $\square$
 646

Corollary 1. Since $|\max_k a_k - \max_k b_k| \leq \max_k |a_k - b_k|$, Lemma 1 and $\|W - D_k\|_F \leq B + \|D_k\|_F$ on $\mathcal{W}_{r,B}$ give the bound on $\sup_{\mathcal{W}_{r,B}} |F - \hat{F}|$. Next, $\mathcal{W}_{r,B} \subseteq \mathcal{W}_r$ implies $\inf_{\mathcal{W}_r} \hat{F} \leq \inf_{\mathcal{W}_{r,B}} \hat{F}$, so the ϵ_{opt} -optimality of $\hat{W}$ over $\mathcal{W}_r$ carries over to $\mathcal{W}_{r,B}$ whenever $\hat{W} \in \mathcal{W}_{r,B}$. Lemma 2 with $\mathcal{W} = \mathcal{W}_{r,B}$ and $\inf_{\mathcal{W}_{r,B}} F \leq F(W^*)$ then completes the proof. The bound applies to our solver’s output only a posteriori, when the chosen B covers both $\hat{W}$ and W^* . The solver itself never enforces a norm bound. $\square$

Lemma 3. Write $q = \text{tr}[(W - D_k)S_k(W - D_k)^\top]$ and let $\hat{q}$ be the same expression with $\hat{S}_k$. Then

$$\frac{q}{d_k} - \frac{\hat{q}}{\hat{d}_k} = \frac{q - \hat{q}}{\hat{d}_k} + \frac{q}{d_k} \cdot \frac{\hat{d}_k - d_k}{\hat{d}_k}.$$

By Lemma 1, $|q - \hat{q}| \leq \|\Delta_k\|_{\text{op}}\|W - D_k\|_F^2$. Moreover $|\hat{d}_k - d_k| = |\text{tr}(D_k \Delta_k D_k^\top)| \leq \|\Delta_k\|_{\text{op}}\|D_k\|_F^2$. Combining the two proves the claim. The assumption $\hat{d}_k \geq d_{\min,k} > 0$ is what keeps the bound finite when it is used uniformly over draws. For a single draw the observed $\hat{d}_k > 0$ suffices. We make no claim about how large $\|\Delta_k\|_{\text{op}}$ is with high probability. $\square$

B IMPLEMENTATION AND NUMERICAL AUDIT

The implementation is pure Python (standard library only), because no numerical library was installed in the execution environment.

- *Inner step.* The inner problem is solved with Proposition 1, using Jacobi eigendecompositions and one-sided Jacobi SVDs.
- *Dual ascent.* It consists of 300 diminishing-step exponentiated-gradient steps followed by 200 backtracking steps. Every evaluated $g(\lambda)$ is a lower bound in exact arithmetic.
- *Primal refinement.* Refinement runs only when the dual gap exceeds 1×10^{-9} in relative terms. It alternates two convex block problems, one with the column space fixed and one with the row space fixed, with smoothed local descent on a factorisation. A candidate is accepted only if it lowers $\hat{F}$.
- *Test suite.* The tests compare the inner solver with a raw-data reduced-rank regression, a grid search, and multi-start alternating least squares. They check the minimax bounds against a grid-plus-ellipsoid reference on $2 \times q$ problems. They reproduce the published fair-PCA gap instance (Tantipongpipat et al., 2019, Lemma 6.2) and the two-group exactness (Tantipongpipat et al., 2019, Thm. 1.2). They also check invariance under $(B, A) \mapsto (BR, R^{-1}A)$.

Passing tests are evidence about the implementation on the tested instances, not proofs.

C ADDITIONAL RESULTS

Table 5 lists the population worst-task error of every stage-2 cell. Table 6 lists all stage-1 methods, including the weight-space baselines.

D NUMERICAL CHECK OF THE BOUNDS

The check was registered before execution (commit 8428292). Stage 2 stored fitted matrices only as SHA-256 hashes, so the 27 matrices ($\hat{W}$ for two arms and 4 draws per teacher, plus W^* per teacher) were regenerated with the stage-2 code path. All 27 hashes match, and the regenerated population values reproduce the recorded ones exactly (largest difference 0.000). This is a re-analysis of recorded instances, not a new experiment.

For Lemma 3 used uniformly on $\mathcal{W}_{r,B}$, we bound $\sup_{\mathcal{W}_{r,B}} E_k(W; S_k, d_k) \leq \|S_k\|_{\text{op}}(B + \|D_k\|_F)^2/d_k$ and use the observed $\hat{d}_k > 0$. The median ratio of left- to right-hand side in the

Table 5: Population worst-task error F for every stage-2 cell, one row per teacher and calibration draw. “orig. (seen)” is the stage-1 draw, whose test results were seen before the diagnostic was designed.

Teacher	Draw	Emp/Emp		Emp/Pop		Pop/Emp	
		Unif.	Minimax	Unif.	Minimax	Unif.	Minimax
0	orig. (seen)	0.657	0.726	0.765	0.712	0.609	0.600
0	r101	0.635	0.550	0.739	0.665	0.632	0.586
0	r102	0.651	0.645	0.628	0.686	0.581	0.533
0	r103	0.584	0.550	0.638	0.754	0.754	0.727
0	Pop/Pop (oracle)	Unif. 0.557, Minimax 0.491					
1	orig. (seen)	0.655	0.768	0.636	0.813	0.719	0.669
1	r101	0.616	0.620	0.634	0.684	0.685	0.642
1	r102	0.699	0.619	0.751	0.647	0.634	0.681
1	r103	0.660	0.693	0.671	0.717	0.646	0.566
1	Pop/Pop (oracle)	Unif. 0.658, Minimax 0.520					
2	orig. (seen)	0.780	1.081	0.736	0.909	0.640	0.757
2	r101	0.694	0.821	0.790	0.816	0.778	0.822
2	r102	0.755	0.726	0.676	0.760	0.742	0.719
2	r103	0.694	0.822	0.761	0.908	0.701	0.648
2	Pop/Pop (oracle)	Unif. 0.673, Minimax 0.597					

Table 6: Stage-1 synthetic fixture: all methods at rank $r = 4$, means over 3 teachers, evaluated on the test sample. †: unofficial simplified re-implementation. ‡: exceeds the rank budget and is shown for reference only. §: factor averaging depends on the factorisation gauge and is a diagnostic only. Weight-space baselines were tuned on the development split, whereas uniform WRRR and minimax have no hyper-parameters.

Method	Final rank	Test worst (mean $\pm$ sd)	Test mean	Cal. worst
Base model ($W = 0$)	0	1.000 $\pm$ 0.000	1.000	1.000
TA + SVD	4	1.445 $\pm$ 0.472	0.770	1.400
KnOTS-TA + trunc.†	4	1.445 $\pm$ 0.472	0.770	1.400
KnOTS-TIES + trunc.†	4	1.077 $\pm$ 0.282	0.770	0.996
CtM-like†	4	1.585 $\pm$ 0.421	0.835	1.543
RegMean (full rank)‡	12	1.341 $\pm$ 0.506	0.701	0.948
RegMean + Eucl. trunc.	4	1.353 $\pm$ 0.414	0.734	1.012
RegMean + whitened trunc.	4	1.339 $\pm$ 0.401	0.739	1.012
Uniform WRRR (rel.)	4	0.685 $\pm$ 0.060	0.588	0.601
Minimax (rel.)	4	0.857 $\pm$ 0.179	0.627	0.472
Factor averaging§	4	1.253 $\pm$ 0.350	0.747	1.204

Table 7: Deterministic bounds evaluated on the recorded fits. Entries are medians with [min, max] over the 3 teachers and 4 calibration draws per teacher. The draws include the original one, which was seen earlier. The bound is $2\delta + \epsilon_{\text{opt}}$ on $\mathcal{W}_{r,B}$ with the a-posteriori radius $B = \max(\|\hat{W}\|_F, \|W^*\|_F)$.

Arm	Cells	$\max_k \ S_k - \hat{S}_k\ _{\text{op}} / \ S_k\ _{\text{op}}$	bound $2\delta + \epsilon$	realised $F(\hat{W}) - F(W^*)$	bound holds
Emp/Pop (Cor. 1)	12	0.73 [0.64, 1.04]	22.50 [14.03, 30.85]	+0.208 [+0.127, +0.311]	12/12
Emp/Emp (Lem. 3)	12	0.73 [0.64, 1.04]	101.42 [69.97, 140.53]	+0.164 [+0.060, +0.484]	12/12

pointwise checks is 0.034. CPU time was 216.1 s under a hard limit of 1100 s, with peak RSS 23.4 MiB.

E CORRECTION HISTORY

Summaries written before this draft contained the following discrepancies with the raw files. The original summaries are preserved in the repository (`CORRECTIONS.md`).

1. The stage-1 minimax mean test worst-task error was summarised as “0.858”. The raw value is 0.857499 (`per_method.minimax_rel.test_worst_rel.mean`), so the correct three-decimal value is 0.857. The earlier value came from rounding a four-decimal display twice.
2. The largest Monte Carlo deviation was summarised as “3.03”. The raw maximum is 3.38; the earlier value was read from the last log line only.
3. The triples 0.713/0.802/1.057 and 0.726/0.768/1.081 were placed side by side in a summary. They are the same minimax solutions evaluated on the test sample and on the population objective, respectively (Table 1). They are not a comparison of two methods.
4. An earlier interpretation said that the minimax solution failed because the weakest task received a small weight. The oracle Pop/Pop solution also gives that task a small weight ($\lambda = 0.165$ and 0.188 for teachers 0 and 2). The small weight itself is therefore not the failure.

F PLANNED EXPERIMENTS (NOT RUN)

None of the following was executed. They are listed so that the absence of results is explicit.

1. **Real-adapter pilot (BLOCKED)**. The plan is to merge 4 public Flan-T5-base GLUE LoRA adapters (rank 16, query/value projections) and to report worst-task and mean task metrics at a matched final rank and matched calibration access. It is blocked for three reasons:

- the adapter repositories contain no license or usage terms;
- the base-model revision used for training is not recorded (the adapter configuration has `revision: null`);
- the training and checkpoint-selection split is undocumented, while the publisher evaluates on the GLUE validation split.

The metric plan must also separate the publisher’s exact-match accuracy from the official GLUE metrics (Section 6). The publisher reports the CoLA adapter at the same CoLA score as the base model (69.1), so accuracy normalised by the adapter’s own score carries no information for that task.

2. **Absolute versus relative errors**, to separate the effect of relative normalisation.
3. **Calibration-size dependence** (not approved in this phase: no n , rank or task-count sweeps).
4. **Regularised worst-task variants**, for example shrinking λ towards uniform. These are not implemented; any such variant must be pre-registered and selected on development data.

G REPRODUCIBILITY DETAILS

- *Stage 1*. Command: `PYTHONPATH=src python3 -m lowrank_merge.run_cpu --config configs/cpu_synthetic.json`. Wall time 58.1 s.
- *Stage 2*. Registered before execution at commit 06947af. Command: `PYTHONPATH=src python3 -m lowrank_merge.run_diag --config configs/p2_stage2_moment_normalizer_diag.json`.
 - 78 of 78 cells completed.
 - CPU time 285.8 s under a hard limit of 1800 s; wall time 288.5 s; peak RSS 46.9 MiB.
 - One worker and one compute thread. No package was installed and nothing was downloaded.

- *Stage 3 (bound check).* Command: `PYTHONPATH=src python3 -m lowrank_merge.run_bound_check --config configs/p2_stage3_bound_check.json`. Wall time 217.6 s.
- *Numbers and tables.* Every number in the text is emitted by `paper/tools/export_results.py`. The file `paper/generated/provenance.tsv` maps each number to its source file and field.